

Software Development Interns Task Plan – Enhanced Task Management System

Project Phase: Feature Expansion, Analytics, Deployment

Duration: Weeks 4–6

Final Submission Date: 25 May, 2026

Week 4 – Collaborative Features & Real-Time Notifications

Objective: Enable multi-user collaboration and implement a notification system for improved team interaction.

Tasks:

1. User Collaboration

Update database schema to support **owner** and **sharedWith** fields (array of user IDs).

- Update API endpoints:

- **PUT /tasks/:id/share** – Share tasks with other users

- **GET /tasks/shared** – Retrieve tasks shared with the user

2. Real-Time Notifications

Use **Socket.IO** or similar library to implement notifications.

- Notify users in real-time when:

- A task is shared with them

- A task's status is updated

- Create **GET /notifications** endpoint for past notifications

3. Frontend Integration

Add “Share Task” button in Task Details component.

- Display notifications in a pop-up or sidebar.

4. Testing & Error Handling

Ensure only authorized users can share/edit tasks.

- Handle invalid sharing attempts gracefully.

Deliverables:

- Real-time notifications working across users.
- Task sharing functionality implemented.
- Updated database schema & API endpoints.
- Code committed & pushed to GitHub.

Week 5 – Advanced Analytics & Reporting

Objective: Provide data-driven insights into task performance through an analytics dashboard.

Tasks:

1. Analytics Dashboard

Create dashboard showing:

- Total tasks (created, completed, pending)
- Weekly/monthly trends (completed vs. overdue)
- Status breakdown (pie chart: Pending, In Progress, Completed)
- Use **Chart.js** or **Recharts** for visualizations.

2. Backend Enhancements

New endpoints:

- `GET /analytics/overview` – Summary stats
- `GET /analytics/trends` – Weekly/monthly trends

3. Frontend Integration

Connect dashboard to dynamic backend data.

- Display charts & graphs with real-time updates.

4. Data Optimization

Use **MongoDB Aggregation Framework** for efficient queries.

- Optimize API responses for faster loading.

Deliverables:

- Fully functional analytics dashboard.
- Backend endpoints delivering analytics data.
- Optimized queries for high performance.
- Code & dashboard screenshots uploaded to GitHub.

Week 6 – Deployment, Documentation & Final Enhancements

Objective: Prepare for production launch with a polished, well-documented, and fully tested system.

Tasks:

1. Deployment

Deploy to **Render**, **Vercel**, or **Heroku**.

- Ensure frontend & backend run on a single live URL.

2. Final Enhancements

Add **dark mode** toggle to the frontend.

- Enable **task attachments** (images, files).
- Improve mobile responsiveness.

3. Documentation

Create a **comprehensive README** including:

- Setup instructions (frontend & backend)
- API documentation with examples
- Screenshots & feature descriptions

4. Testing

Perform end-to-end testing.

- Fix all identified bugs before submission.

Deliverables:

- Fully deployed live project URL.

- README with complete documentation.
- Dark mode & attachments implemented.
- Video walkthrough of the deployed app.

Evaluation Criteria

1. **Feature Implementation** – Task sharing, notifications, analytics, dark mode, attachments.
2. **Performance** – Optimized backend, responsive frontend, fast API calls.
3. **Documentation** – Clear README, well-structured code.
4. **Deployment** – Smooth live deployment accessible via one link.

Final Deadline: 25 May, 2026– 11:59 PM PKT

All deliverables must be submitted via GitHub, along with the live URL and video walkthrough.